

EXPERIENCE

- **Citadel (Fixed Income and Macro)** London, United Kingdom
Senior Software Engineer → Engineering Manager *April 2023 - Present*
 - **Research Data Platform (Storage & API Layer):**
 - * Sole architect and developer of a centralized research data platform, now adopted by **all systematic desks**
 - * Built on **PostgreSQL and Arrow Flight** over gRPC with bidirectional streaming APIs directly from database cursors; deployed on **Kubernetes**
 - * Introduced sharded per-desk architecture for independent scalability, enabling heavy-usage teams to run their own stacks, with built-in permissions and TTLs to protect desk-specific IP and restrict the growth of unused data
 - * Designed and implemented the research storage API, which became the preferred interface for department-wide data access, prompting migration of other storage systems to use this API due to its intuitive design and streamlined workflow
 - **Market Data Infrastructure (Tick Pipeline):**
 - * Sole developer of a tick data ingestion pipeline ingesting data from multiple market venues in real time, processing **30B+ market events/day** across **27M+ unique instruments**
 - * Built a low-latency, high-throughput system using **Kafka, Redis, and C++** with concurrent and lock-free design patterns; **primary expertise is in system design and Python-based data engineering**
 - * Designed for production resilience, scalable for increased market data needs, and seamlessly rolled out without disruptions to existing usecases
 - **Leadership & Platform Transformation:**
 - * **Promoted to Engineering Manager**, leading two teams (6 engineers) focused on data onboarding and data platform infrastructure
 - * Modernized the data onboarding platform, overseeing migration of **5,000+ ETL jobs** and reducing onboarding times from 5+ months to **<30 days**
 - * Architecting and leading the team developing a department-wide data quality and discovery platform, providing validation checks, dataset health monitoring, metadata management, multi-storage support, and data arbitration
- **University of St. Andrews** United Kingdom
Career break for full-time study (details in Education section) *September 2021 - September 2022*
- **Bloomberg LP** London, United Kingdom
Senior Software Engineer *2019 - September 2021*
 - **Taxonomy and Metadata System:** Infrastructure team developing a taxonomy and metadata system to support various asset classes at Bloomberg. Python, GraphQL, Redis, RabbitMQ, Solr, RDF, PostgreSQL, Data Pipeline
 - * **Architected ingestion pipelines** that handle ingestion of metadata for datasets such as commodity index tickers.
 - * Added **support for sharding** to increase the speed of the ingestion pipeline. Split up the initial ingestion into 8 shards that are running distributed across an entire cluster.
 - * Implemented node overlay and concatenation executed on RDF graphs.
 - * Worked on **rearchitecting the ingestion pipeline** from a synchronous pipeline to a file based ingestion.
 - * Implemented services in GraphQL to avoid overfetching and underfetching on the existing Bloomberg specific microservice framework.
 - * Worked on a side project, debugging tool, to help find metadata for given entities regardless of the source of provenience, using a **topological sort algorithm on service APIs and RDF Graphs**.
 - * Team lead backup for a team of 7 engineers.
- *Software Engineer* *2016 - 2019*
 - **Real Time Analytics:** Application team, maintaining a large variety of analytics, integrated in different products across Bloomberg. Processing several billion trades per day. C++, Distributed, Realtime
 - * Implemented Average Volume at Time for non equities and Top Exchanges in the past X minutes.
 - * Implemented a **load testing facility** that is used to simulate the impact of market open on the system to assess performance and capacity. I identified a bottleneck in the system and by fixing it, subscriptions are being processed 3 times faster.

- * **Increased availability** of an analytic by implementing fallback to disk in case the database connection cannot be established. This prevents the 1 minute downtime per week when the database cluster is being rebooted.
- **Recommendation Disclosures and Trade Ideas:** Python, Typescript, Microservices, Event Sourcing
 - * Implemented asynchronous messaging using RabbitMQ to allow distribution of recommendations. Added support to publish messages, implemented consumers and set up the middleware.
 - * Reduced the codebase by 8% by removing around 100.000 lines of dead or unused code.

Intelligent Software Systems

Timisoara, Romania

Software Engineer

2013 - 2016

- Built a desktop-to-CRM sync tool (workbooks.com) and contributed features and bug fixes to the FAFSA questionnaire.

EDUCATION

University of St. Andrews

United Kingdom

MSc Statistics

Distinction, Deans' List, grade 17.9/20, September 2022

- **Modules:** Markov Chains, Bayesian Statistics, Multivariate Analysis, Statistical Modelling using GLM/GAMs, Quantitative risk analysis, Machine Learning
- **Dissertation:** An Ensemble of HMM and KNN for signal generation in horse betting markets. Grade 19.3/20 - Highest grade in my cohort

University of Sheffield

Remote, United Kingdom

Graduate Certificate in Statistics

Distinction, grade 90/100, July 2021

- **Modules:** Mathematical Methods for Statistics, Probability and Probability Distributions, Basic Statistics

Politechnica University of Timisoara

Timisoara, Romania

B.S.E in Computer Science

July, 2014

- **Modules:** Software Engineering, Computer Architecture, Algorithms, Software Development Fundamentals, Design Patterns, Programming Languages, Databases

SKILLS

Coding

- **attrs-strict** <https://github.com/bloomberg/attrs-strict>: Created and open-sourced a library that allows runtime validation for attrs dataclasses in Python.
- **bloomberg-bas-middleware**: Maintainer and creator of an inner source library used for Python services at Bloomberg. Created Flask-like middlewares that are used in over 10.000 services running in Python 3 at Bloomberg. Any new service has by default the functionality opted-in.
- **Python Guild**: Part of a group of 18 engineers leading the direction and adoption of Python at Bloomberg, led the Design Reviews Working group, helping teams to bring projects to a broader audience.

Public speaking for large audiences

- **"How to migrate services to Python 3"** - Attendance of over 200 engineers.
- **"Modeling the commodity world"** - Presentation on the architecture of the system I worked on and the different architectural decisions taken in the past years. Attendance of over 150 engineers.

Mentorship

- Mentored interns and junior/mid engineers at Citadel and Bloomberg, conducted interviews, ran workshops on Python 3 migration, and supported new joiners in ramping up and contributing to inner-source projects.

Architecture and technology

Large scale distributed systems, Data pipelines, Microservices, GraphQL, PostgreSQL, Redis, Kafka, Solr, RabbitMQ, Event Sourcing, Git, RDF

Data onboarding

External data ingestion via APIs, SFTP, web scraping and PDF parsing; sources include Bloomberg (PerSecurity/BLPAPI), Refinitiv, S&P; ETL pipeline management at scale

Languages

Python (primary), C++ (production use), JavaScript/TypeScript (basic), SQL

Mathematics

Probability theory, Markov chains, Bayesian models, GLMs, statistical inference